

NUMIC — Overview for stakeholders

Clinical decision support for post-haemorrhagic ventricular dilation (PHVD) risk

Audience: Clinical, regulatory, and quality leadership (e.g. medical device QMS)

Purpose: Single-page summary of intent, design, and responsible next steps — not a regulatory submission.

1. summary

NUMIC is a software concept for **structured assessment and communication of risk** in infants with **post-haemorrhagic ventricular dilation (PHVD)** after intraventricular haemorrhage. It combines:

- **Standard cUS measurements** used in practice and trials (ventricular index, anterior horn width, thalamo-occipital distance),
- **Explicit, versioned scoring rules** aligned with published thresholds and trial logic (e.g. ELVIS-era framing),
- **Temporal (“progression”) information** — how fast ventricular size is changing between scans — which the literature highlights as prognostically important but often **not quantified in a single, actionable score**.

The goal is **decision support**: to make **size + trajectory + key clinical context** visible in one place, with **transparent sub-scores** and **tiered suggested actions** (e.g. surveillance vs escalation), so neonatal teams can align discussion and documentation — **not** to replace clinical judgment or to make an autonomous diagnosis.

2. Clinical problem

PHVD after IVH is associated with important morbidity. Management depends on **how dilated the ventricles are, how quickly dilation progresses, and clinical evolution**. Teams already use serial cranial ultrasound (cUS) and published cut-offs; however:

- Progression is often **described qualitatively** (“a bit bigger”) rather than **summarised in a consistent, auditable way**.
- There is **no single, widely adopted composite** that combines **absolute size, rate of change, and late clinical signs** for handover and quality review.

NUMIC is intended to **support** that gap: a **structured, reproducible summary** that still maps back to familiar metrics.

3. What we are building (product intent)

Layer	Role
Measurement capture	Accept VI, AHW, TOD (and related fields) from PACS/overlay pipelines, image-derived tools, manual/GUI entry, or bulk import — with patient identity, date/time of study, who measured, and clinical notes where available.
NumicFlow scoring	Static score from current size vs thresholds; progression score from change vs a prior timepoint; clinical modifier from structured concern level (e.g. fontanel / systemic signs), with rules versioned (e.g. <code>numic_flow_v1</code>) so outputs remain interpretable when policies evolve.
Risk tier	Map total score to low / moderate / high bands with plain-language suggested actions (aligned to published pathways — scan frequency, escalation, neurosurgical awareness).
Presentation	Eventually: clinician-facing UI (not raw API JSON) — patient context, numbers, trajectory plot, score breakdown, rule version, and export for records.

A **demo API** exists to show **end-to-end behaviour** for prototyping; the **clinical product** would be a regulated front-end and deployment model, not the demo route alone.

4. NumicFlow — high level

Static layer (0–6)

Per-metric bands for VI (with percentile / nomogram logic as implemented), AHW, and TOD, consistent with the evidence base we are coding against.

Progression layer (0–6)

Quantified change in VI, AHW, TOD between **current and prior** measurements over time (worsening-direction logic as specified in our rule set).

Clinical modifier (0–2)

Structured input for **late but important** clinical concern (none / mild / clear), from bedside observation or file — not a full neonatal exam replacement.

Total

Weighted sum → **NumicFlow score (0–14)** → **risk tier** with **action bands**.

All thresholds live in **named, versioned rule bundles** so that governance can approve changes and old results remain attributable to a **specific rule version**.

5. System Summary

- **Backend:** API-first service (e.g. FastAPI) separating **measurement ingestion, scoring,** and (later) **persistence**.
- **Data model:** Patient/study context + measurements + optional progression and reports — designed for **audit** (who, when, which rule version).
- **Inputs:** Designed for **hospital reality:** DICOM/PACS metadata, spreadsheet import, manual entry from a measurement GUI, future ML-based image extraction (currently stubbed where models are not yet integrated).

Engineering detail is available separately; this document focuses on **intent and governance**.

6. Quality, safety, and regulatory posture - discussion points

The following are **topics for informed conversation with QMS / clinical governance**, not conclusions:

- **Intended use:** Decision support / workflow aid vs diagnostic device claims — must be nailed down with **intended user, indication,** and **contraindications** in a formal definition.
- **Clinical validation:** Agreement on **performance study design,** reference standard, and neonatology sign-off before any deployment near patients.
- **Risk management:** ISO 14971-style hazards (wrong input, wrong rule version, UI misread, missed escalation) and mitigations (human-in-the-loop, versioning, logging).
- **Software lifecycle:** IEC 62304 class, configuration management, release records, and **SBOM** / cybersecurity expectations for a connected product.
- **Data protection:** UK GDPR / NHS DSPT-style considerations if patient identifiers are processed.
- **Human factors:** UI testing so **risk tier** and **recommended actions** are unambiguous and match local protocol.

NUMIC is at an early stage: **architecture and rule structure exist in software; formal QMS alignment and clinical evidence** are deliberate next steps — which is why engagement with UCL quality leadership is valuable **before** scaling claims or deployment.

7. Current status vs roadmap

Stage	Status (high level)
Scoring logic & versioning	Prototype implemented; requires clinical validation and change control under QMS if marketed.
Measurement ingestion	Today (prototype) : manual entry, spreadsheet import, and numeric feeds from overlay/PACS-style metadata. Roadmap : deeper PACS/workflow integration where needed; eventually, automatic VI/AHW/TOD from cUS images using segmentation models , with clinical validation before use in care.
Clinician UI	Roadmap : develop a clinician-facing UI focused on ease of use —the production experience, not raw APIs or demo pages. Input from clinicians (neonatologists, sonographers, and related roles) will be essential for screen flow, terminology, and how risk tiers and suggested actions are presented; the API and demo remain for engineering and early testing.
Regulatory strategy	Open — depends on intended use, jurisdiction (UKCA / MDR), and whether the system is Software as a Medical Device and in which class.

8. What we would value from this conversation

NUMIC's next steps depend heavily on **how** we position the product under device law and **what** evidence and governance UCL or partner sites would expect. Your perspective—drawing on device QMS and clinical governance—would help me **prioritise** documentation, timelines, and who to engage next.

- **Reality check** on positioning NUMIC as **clinical decision support** under UK / EU device thinking.
 - **Minimum viable QMS path** if we pursue **clinical pilots** or **UCL-affiliated evaluation**.
 - **Clinical evaluation & risk management — where to start**: How far must we go (plan vs evidence) before showing **risk categories** or **suggested actions** to clinicians outside a purely internal prototype? Which artifacts you'd expect early on (e.g. intended use wording, clinical benefits/risks narrative, usability/safety considerations, post-market thinking) and **what level of formality** is proportionate at pilot stage vs a full **Software as a Medical Device** path.
 - **Institutional gate — what to line up first**: Your read on **UCL** (or typical NHS trust) expectations before **any** real patient data or clinician-facing use: e.g. **DPIA**, information governance, **research vs service / clinical evaluation** boundaries, and whether we should treat an evaluation as “research” with full ethics first, or a defined service improvement path. **Concrete ask**: what we should **secure in writing** and **in what order** so we do not block useful pilots or create avoidable compliance debt.
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9. NUMIC – one line overview

“A structured PHVD risk summary that combines serial cUS size, rate of change, and key clinical modifiers into a transparent, versioned score — intended as decision support for neonatal teams, with formal device and clinical pathways still to be defined with quality and clinical leads.”

References

1. El-Dib M, Limbrick DD Jr, Inder T, Whitelaw A, Kulkarni AV, Warf B, Volpe JJ, de Vries LS. Management of Post-hemorrhagic Ventricular Dilatation in the Infant Born Preterm. *J Pediatr*. 2020 Nov;226:16-27.e3. doi: [10.1016/j.jpeds.2020.07.079](https://doi.org/10.1016/j.jpeds.2020.07.079). Epub 2020 Jul 30. PMID: 32739263; PMCID: PMC8297821.
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